

# Software Requirement Specification

COS 301 - Capstone 2026  
Capstone Project: EquityLens  
By The Big Five



Project Name: EquityLens  
Client: Amazon Web Services (AWS)  
Team: The Big Five (TB5)

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# 1. Introduction

## 1.1 Purpose

This document specifies the functional and non-functional requirements for EquityLens, an investment analytics platform. It is intended for the development team (The Big Five - TB5), project examiners and the stakeholders at AWS to define the scope and success criteria for the project. It is written so that a developer with no prior knowledge of the codebase could implement the system from the requirements without being constrained to specific technologies.

## 1.2 Business Need

Retail investing in South Africa has surged due to accessible brokerage platforms, but a significant analytical gap remains, this gap entails retail investors having access to the market yet lack the tools needed to interpret risk metrics and connect world events to portfolio volatility. Investors face data overwhelm - they hold diversified portfolios but cannot easily calculate total risk exposure, or understand why specific stocks moved, without manual and error-prone spreadsheets.

EquityLens addresses this by centralising investment data into a single dashboard. It automates the calculation of financial indicators, correlates news events with price movements, and uses a conversational AI assistant to explain metrics in plain language - bridging the gap between raw data and actionable insight, at no cost to the user.

## 1.3 Scope

EquityLens is a web-based analytics suite for South African retail investors. It covers:

- Secure user registration, email verification, multi-factor authentication and login
- Portfolio ingestion via spreadsheet upload (PDF/Excel), brokerage PDF statement upload, or manual entry
- Automated financial indicator calculation with plain-language explanations
- News-to-price-movement correlation for the user's holdings
- A conversational AI assistant with persistent conversation history and safety guardrails
- A dashboard summarising portfolio value, health, performance against a benchmark, and recommended actions
- A watchlist for stocks the user does not yet own

Out of scope: direct stock trading, brokerage integration that requires the user's brokerage credentials, and regulated financial advice.

## 1.4 Definitions, Acronyms and Abbreviations

- EquityLens - the investment analytics web platform described in this document
- SRS - Software Requirements Specification
- SAS - Software Architecture Specification

- FR - Functional Requirement
- NFR - Non-Functional Requirement
- UC - Use Case
- US - User Story
- TUCBW - This Use Case Begins When
- TUCEW - This Use Case Ends When
- MFA - Multi-Factor Authentication
- TOTP - Time-based One-Time Password
- API - Application Programming Interface
- AWS - Amazon Web Services
- POPIA - Protection of Personal Information Act (South Africa)
- JSE - Johannesburg Stock Exchange
- ALSI - All Share Index (JSE)
- LLM - Large Language Model
- ETF - Exchange Traded Fund
- CAPM - Capital Asset Pricing Model
- RSI - Relative Strength Index
- P/E - Price-to-Earnings ratio
- CI/CD - Continuous Integration / Continuous Deployment

## 1.5 Document Overview

Section 2 describes the intended users and their goals as user stories. Section 3 presents the use case model: a use case diagram followed by a TUCBW/TUCEW description of each use case. Section 4 specifies the functional requirements, assigned to eight subsystems. Section 5 quantifies the five targeted quality requirements. Section 6 presents the domain model as a UML class diagram with entity descriptions. Section 7 traces each use case to the functional requirements that realise it, and Section 8 records the assumptions and external dependencies the requirements rest on. How the system is structured to satisfy these requirements is specified separately in the Software Architecture Specification (SAS).

## 2. User Stories and User Characteristics

### 2.1 Intended Users

Retail Investor - the primary and only in-app user. A South African retail investor who imports portfolio data and uses the analytics, news, AI assistant and watchlist features. Assumed to have basic familiarity with stock tickers but no formal finance training therefore everything must be explainable in plain language.

EquityLens deliberately has a single in-app role. System administration (account management, configuration, monitoring) is an operational responsibility carried out by the development team through the hosting platform's own tools and it is not a user-facing capability of the system and no administrative interface is in scope.

### 2.2 User Stories

#### 2.2.1 Authentication

- US-1: As a new user, I want to create an account so that my portfolio data is saved securely.
- US-2: As a new user, I want to verify my email address with a code so that only I can activate my account.
- US-3: As a security-conscious user, I want to protect my account with an authenticator app code so that a stolen password alone cannot access my data.
- US-4: As a returning user, I want to log in so that my session remains active while I work.
- US-5: As a user on a shared device, I want to log out so that others cannot access my data.

#### 2.2.2 Portfolio Management

- US-6: As an investor, I want to upload a spreadsheet of my holdings so that I don't have to enter them manually.
- US-7: As an EasyEquities user, I want to upload my PDF statement so the system can extract my holdings automatically.
- US-8: As a user with a small portfolio, I want to manually add a holding so I can track new investments immediately.
- US-9: As a cautious user, I want to preview what the system extracted from my file before it is saved so that mistakes don't corrupt my portfolio.

#### 2.2.3 Analytics and Financial Indicators

- US-10: As an investor, I want to see my portfolio's Sharpe Ratio so that I can judge whether my returns justify the risk.
- US-11: As a novice, I want a plain-language explanation next to every indicator so I can learn what the numbers mean.
- US-12: As a technical user, I want to see the RSI for my stocks so I can identify overbought or oversold conditions.

#### 2.2.4 AI Assistant.

- US-14: As a novice user, I want my AI assistant to have a contextual understanding of my portfolio for my questions.
- US-15: As a novice, I want to ask “Is my portfolio diversified?” and get a natural-language answer grounded in my actual holdings.
- US-16: As a returning user, I want my previous AI conversations saved so I can pick up where I left off.
- US-17: As a user, I want to rename or delete old conversations so my chat history stays organised.

#### 2.2.5 Dashboard and Insights

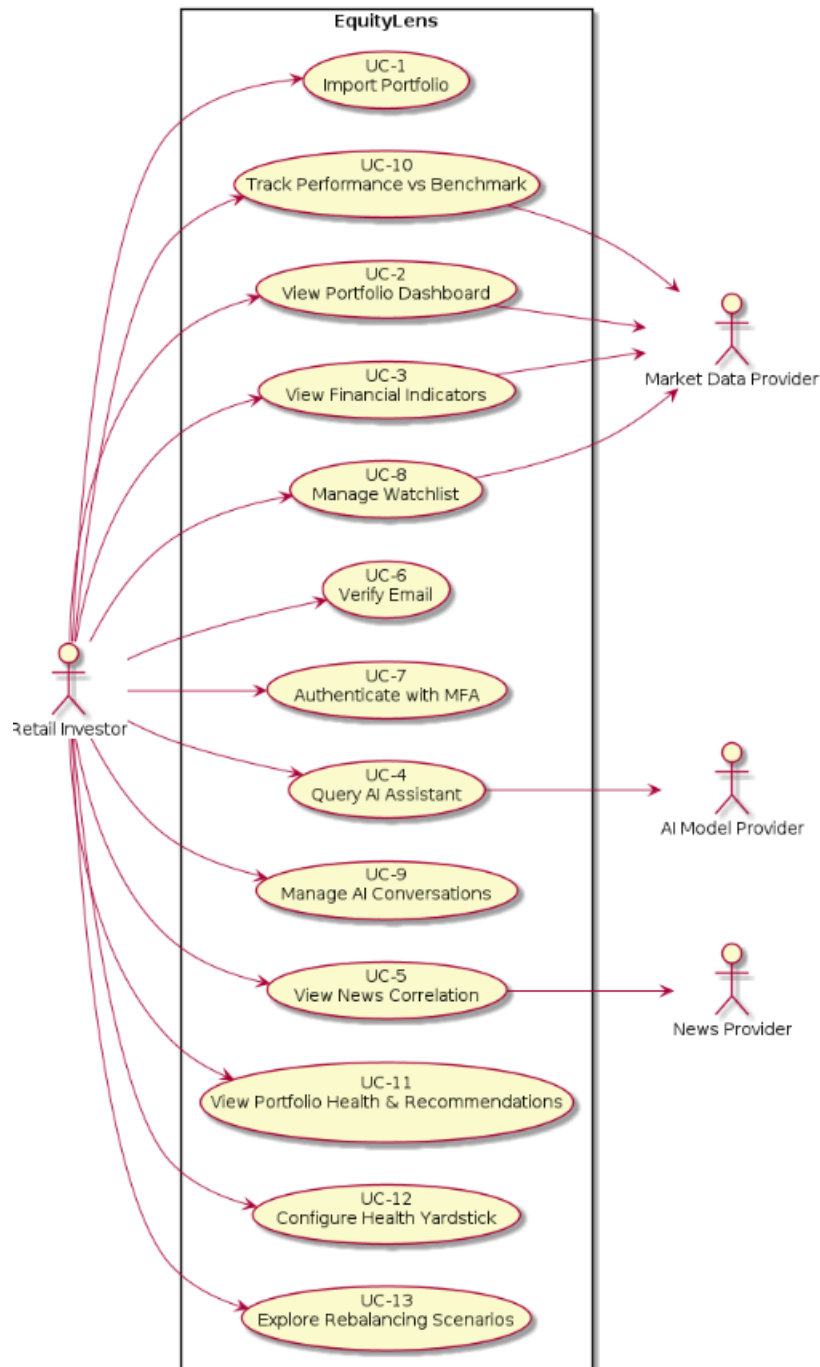
- US-18: As an investor, I want a single health score for my portfolio so I can tell at a glance whether anything needs attention.
- US-19: As an investor, I want to see my performance charted against a benchmark matched to my portfolio's exposed region so I know if I'm beating the market that matters to my portfolio.
- US-20: As a busy user, I want the system to tell me what needs my attention (concentration risk, poor diversification) rather than making me hunt through data.
- US-21: As an investor, I want to see which holdings drove today's gain or loss so I know why my portfolio moved.
- US-22: As an investor, I want to tailor my portfolio health grading based on my goal.

#### 2.2.6 Watchlist

- US-23: As an investor, I want to add stocks I don't yet own to a watchlist so I can track them before buying.
- US-24: As a user, I want to remove a stock from my watchlist when I'm no longer interested.

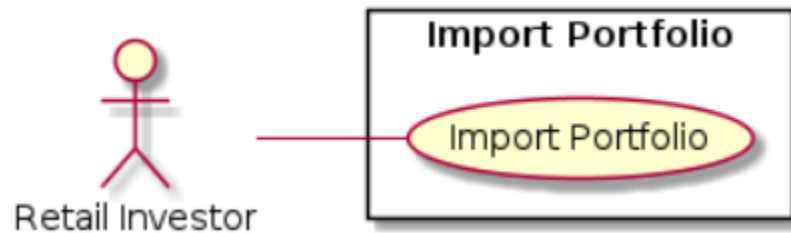
### 3. Use Cases and Use Case Diagram

#### 3.1 Use Case Diagram



## 3.2 Use Case Descriptions

### UC-1: Import Portfolio



Actor: Retail Investor

Precondition: The user is authenticated and on the Portfolio page. The user has a portfolio file (PDF statement, Excel) or details needed for manual entry.

Postcondition: The user's holdings are saved against their account and visible in the holdings table.

TUCBW the user clicks the Import button on the Portfolio page.

Main flow:

1. The system displays the upload window.
2. The user selects an Excel workbook that follows the provided template.
3. The system extracts the holdings from the file.
4. The system validates the extracted content.
5. The system displays a preview of the parsed holdings for confirmation.
6. The user confirms the preview.
7. The system saves the holdings and displays them in the holdings table.

Alternative flows:

- PDF statement upload: at step 2 the user selects a brokerage PDF statement instead; the system extracts holdings, transactions, dividends and statement metadata, and the flow continues from step 3.

Exception flows:

- Malformed file: validation at step 4 fails. The system rejects the upload, shows an error message, and keeps the upload window open so the user can retry.
- No content found: extraction succeeds but no holdings are found. The system informs the user and prompts them to enter holdings manually or upload a different file.

TUCEW the user sees their imported holdings displayed in the holdings table.

## UC-2: View Portfolio Dashboard



Actor: Retail Investor

Precondition: The user is authenticated.

Postcondition: The dashboard reflects the user's current portfolio state.

TUCBW the user navigates to the Dashboard page.

Main flow:

1. The system confirms the user's session is valid.
2. The system retrieves the user's holdings, current market values, sector breakdown, performance history and dividend history.
3. The system displays the portfolio value, today's change, the largest positions, sector allocation, and the performance chart.

Alternative flow:

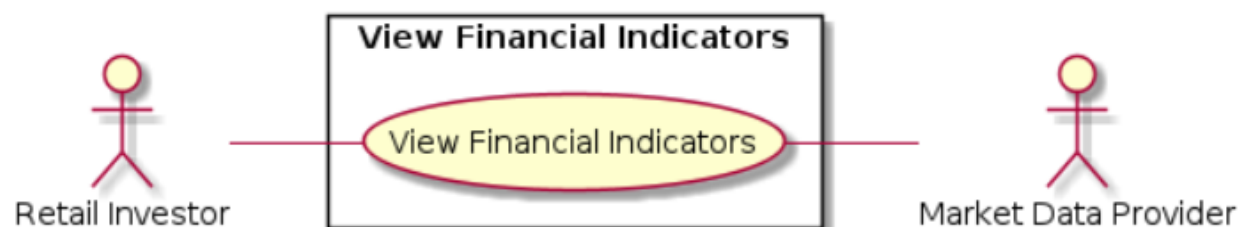
- Empty portfolio: at step 2 no holdings exist. The dashboard displays an onboarding state prompting the user to import a portfolio, with a direct link to the Portfolio page.

Exception flows:

- Unauthenticated user: the session is invalid or missing; the system redirects the user to the login page and the dashboard is not displayed.
- Data retrieval failure: at step 2 the system cannot load the portfolio; the dashboard displays an error state with a retry prompt.

TUCEW the user sees their populated dashboard (or the onboarding state for an empty portfolio).

## UC-3: View Financial Indicators



Actor: Retail Investor

Precondition: The user is authenticated and has at least one holding. Sufficient price history exists to compute each indicator.

Postcondition: An indicator card is displayed for each of the seven supported indicators.

TUCBW the user navigates to the Analytics page.

Main flow:

1. The system retrieves the user's saved holdings.
2. The system calculates the seven supported indicators (Sharpe Ratio, Sortino Ratio, P/E Ratio, CAPM expected return, Altman Z-Score, Beta, RSI) for the relevant holdings.
3. The system displays one card per indicator showing its name, value, and classification where applicable.

Alternative flow:

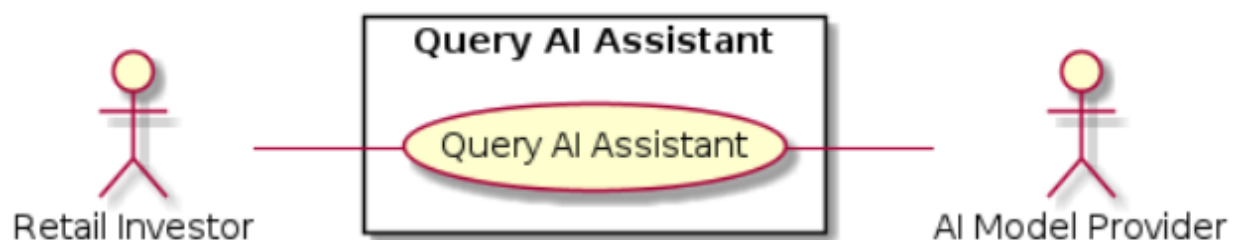
- View explanation: the user hovers over (or focuses) an indicator name; the system displays a plain-language explanation. Moving away hides it without affecting the page.
- Edit indicators: the user presses the "edit" button for a specific stock; the system displays a popup with all the indicators, built-in presets and adding presets functionality.

Exception flows:

- Insufficient data: an indicator cannot be calculated for a holding. Its card shows an N/A state with a short explanation; all other cards render normally.
- Calculation failure: a calculation fails. The affected card shows an error state with a retry prompt; the rest of the page remains usable.

TUCEW the user sees the indicator cards for their portfolio.

#### UC-4: Query AI Assistant



Actor: Retail Investor

Precondition: The user is authenticated and on the AI Assistant page.

Postcondition: The assistant's response is displayed and the exchange is stored in the active conversation.

TUCBW the user submits a typed question in the chat input field.

Main flow:

1. The system sends the question, together with the user's portfolio context, to the AI layer.
2. The AI layer generates a response grounded in the user's holdings, indicators and relevant market data.
3. The system displays the response beneath the user's message, with a disclaimer that responses are informational only.

Alternative flow:

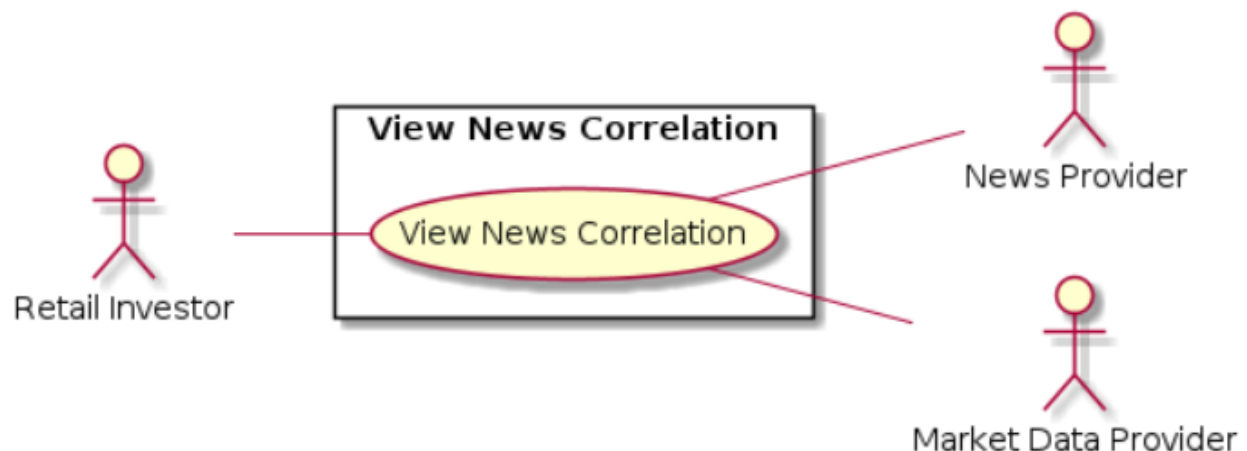
- Card question: the user clicks the assistant trigger on a dashboard card; the assistant opens with that question already filled in, and the flow continues from step 1.

Exception flows:

- Guardrail triggered: the question asks for an explicit buy or sell recommendation. The system responds that EquityLens provides analysis and information only, and directs the user to the Analytics page to inform their own decision.
- AI service unavailable: the primary AI layer fails. The system falls back to the secondary layer automatically. If both are unavailable, an error message prompts the user to try again.

TUCEW the user sees the assistant's response in the conversation.

#### UC-5: View News Correlation



Actor: Retail Investor

Precondition: The user is authenticated and on the News page.

Postcondition: News articles relevant to the user's holdings are displayed, filterable by ticker and date.

TUCBW the user navigates to the News page.

Main flow:

1. The system loads articles relevant to the user's holdings.
2. The system displays each article with headline, source, publication time, related ticker badge, and a price-movement indicator showing how the related stock moved around publication.
3. The user clicks an article; the full article opens in a new tab at the source publication.

Alternative flows:

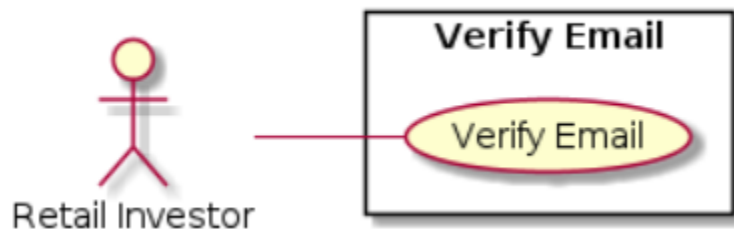
- Filter by ticker: the user clicks a ticker chip; the list filters to articles correlated with that holding.
- Filter by date range: the user selects Today, This Week, This Month, or a custom range; the list filters to that window.

Exception flows:

- No articles found: an empty state prompts the user to broaden the filter or check back later.
- News service unavailable: an error message indicates news is temporarily unavailable and prompts the user to try again.

TUCEW the user sees the news list relevant to their filters.

#### UC-6: Verify Email



Actor: Retail Investor

Precondition: The user has just registered an account and received a 6-digit verification code by email.

Postcondition: The account is activated and the user can log in.

TUCBW the user opens the email verification page after registering.

Main flow:

1. The system displays the verification form, pre-filled with the user's email address where available.
2. The user enters the 6-digit code from their email.
3. The user submits the code.

4. The system confirms the code is correct and activates the account.
5. The system displays a success message and redirects the user to the login page.

Alternative flow:

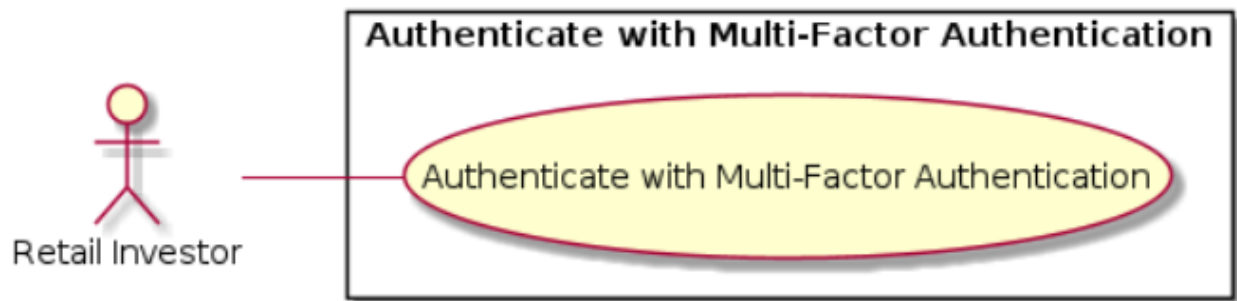
- Arriving without a pre-filled email: the user reaches the page directly; the system additionally asks for the email address before the code can be submitted.

Exception flow:

- Invalid or expired code: the system rejects the code, displays an error message, clears the code field, and allows the user to retry.

TUCEW the user sees the success confirmation and is redirected to the login page.

#### UC-7: Authenticate with Multi-Factor Authentication



Actor: Retail Investor

Precondition: The user has a verified account and is on the login page.

Postcondition: The user is authenticated with a second factor and lands on the dashboard.

TUCBW the user submits a correct email and password on the login page.

Main flow (returning user with MFA configured):

1. The system prompts for a 6-digit code from the user's authenticator app.
2. The user enters the code and submits.
3. The system verifies the code and completes authentication.
4. The system redirects the user to the dashboard.

Alternative flow (first login - MFA setup):

1. The system determines that MFA has not yet been configured and displays a QR code.
2. The user scans the QR code with an authenticator app (or reveals and enters the setup key manually).
3. The user enters the 6-digit code the app generates, to prove the pairing works.

4. The system activates MFA for the account, completes authentication and redirects the user to the dashboard.

Exception flow:

- Invalid code: the system rejects the code, displays an error message, clears the input and allows the user to retry.

TUCEW the user lands on their dashboard, authenticated with both factors.

#### UC-8: Manage Watchlist



Actor: Retail Investor

Precondition: The user is authenticated.

Postcondition: The user's watchlist reflects the additions or removals made.

TUCBW the user opens the watchlist and chooses to add or remove a stock.

Main flow (add):

1. The user searches for a stock by ticker or company name.
2. The system displays matching stocks.
3. The user selects a stock to add.
4. The system saves it to the user's watchlist and displays it with its current price and daily percentage change.

Alternative flow (remove):

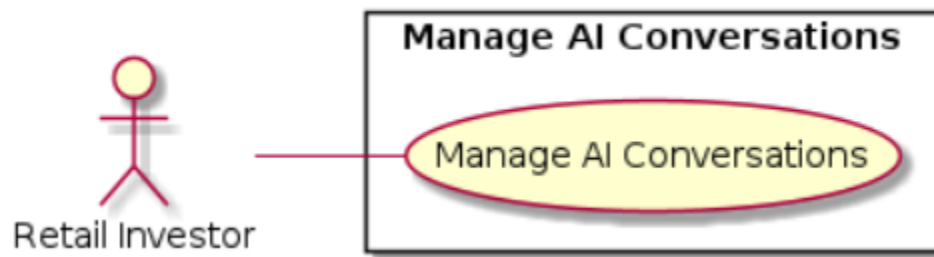
1. The user selects the remove action on a watchlist entry.
2. The system removes the entry and updates the displayed list.

Exception flows:

- No match found: the search returns no stocks; the system shows an empty result state and the user may refine the search.
- Price unavailable: a watchlist entry's market data cannot be fetched; the entry renders with its last known data or a clear unavailable state, without breaking the rest of the list.

TUCEW the user sees their updated watchlist.

## UC-9: Manage AI Conversations



Actor: Retail Investor

Precondition: The user is authenticated and on the AI Assistant page.

Postcondition: The user's conversation list reflects the changes made, and the selected conversation's history is displayed.

TUCBW the user interacts with the conversation list in the AI Assistant sidebar.

Main flow (resume a conversation):

1. The system displays the user's saved conversations.
2. The user selects a conversation.
3. The system loads and displays that conversation's full message history.
4. The user continues the conversation from where it left off.

Alternative flows:

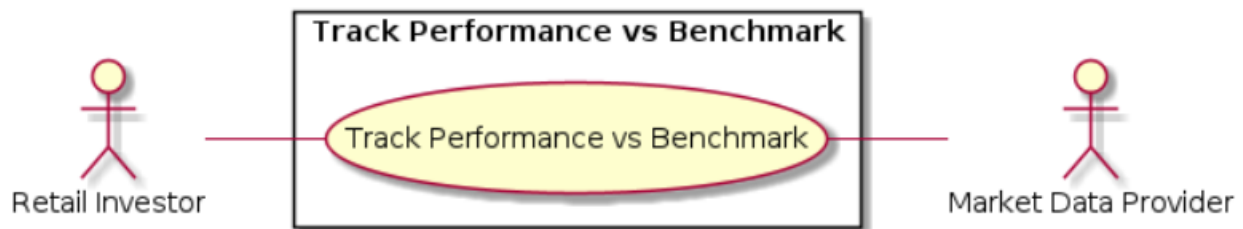
- New conversation: the user clicks New Chat; the system clears the message area and the next question starts a new saved conversation.
- Rename: the user chooses rename on a conversation, types a new title and confirms; the system saves and displays the new title.
- Delete: the user chooses delete on a conversation; the system removes it from the list, and if it was open, clears the message area.

Exception flow:

- History unavailable: the conversation list or messages cannot be loaded; the system shows the page in its empty state and the user can still start a new conversation.

TUCEW the user sees the conversation list and message area reflecting their action.

## UC-10: Track Performance vs Benchmark



Actor: Retail Investor

Precondition: The user is authenticated and has holdings. At least one daily portfolio value snapshot exists.

Postcondition: The user has seen their portfolio's return charted against the benchmark level" with "a benchmark weighted to the portfolio's own regional exposure.

TUCBW the user views the performance chart on the Dashboard.

Main flow:

1. The system records a snapshot of the portfolio's total value and the benchmark level for the current day (at most one per day).
2. The system retrieves the accumulated snapshot history.
3. The system displays the portfolio value and benchmark as two series on one chart, together with the portfolio return, benchmark return, and the difference over the visible window.

Alternative flow:

- Missing benchmark data: the benchmark level could not be fetched on a given day; the chart shows a gap in the benchmark series for that day rather than fabricating a value.

Exception flow:

- No history yet: fewer than two snapshots exist; the chart area explains that performance history will appear once the portfolio has accumulated a few days of data.

TUCEW the user sees their performance plotted against the benchmark with the comparative return figures.

## UC-11: View Portfolio Health and Recommendations



Actor: Retail Investor

Precondition: The user is authenticated.

Postcondition: The user has seen their portfolio health score and any recommended actions.

TUCBW the user views the health score and Action Centre on the Dashboard.

Main flow:

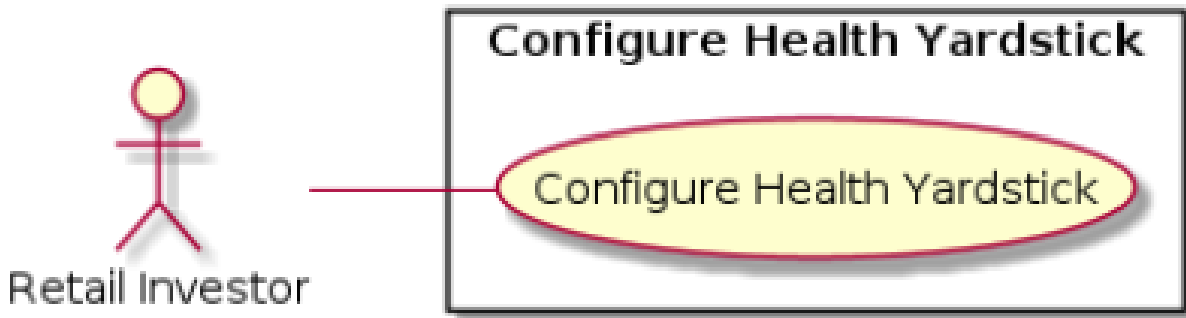
1. The system computes a health score out of 10 from the user's actual holdings, as a weighted combination of sector concentration, single-position concentration and effective breadth. The weights and thresholds come from the active yardstick.
2. The system displays the score, its classification (Excellent / Healthy / Mixed / Needs attention), and the strongest and weakest contributing factors in plain language.
3. The system evaluates the portfolio against defined risk thresholds (sector concentration, single-holding concentration, low diversification) and lists an actionable item for each threshold crossed, each with a link to the relevant page or section.

Alternative flows:

- Nothing to flag: no thresholds are crossed; the Action Centre displays a reassuring all-clear state instead of an empty card.
- No holdings: the health score and Action Centre display an onboarding prompt to import a portfolio.

TUCEW the user has seen their health score and knows what, if anything, needs their attention.

## UC-12: Configure Health Yardstick



Actor: Retail Investor

Precondition: The user is authenticated and viewing the portfolio health card.

Postcondition: The health score is expressed against the selected yardstick, and the choice persists for the user's future sessions.

TUCBW the user clicks Change on the portfolio health card.

Main flow:

1. The system displays the available strategy presets with a short description of what each one optimises for.
2. The user selects a preset.
3. The system saves the choice against the user's account and recomputes the health score using that preset's weights and thresholds.
4. The system redisplay the score, its classification and its contributing factors against the new yardstick.

Alternative flows:

- Revert to the goal-matched yardstick: the user clears their override; the system falls back to the preset derived from their stated goal horizon.
- Reset to default: the user restores the EquityLens default preset.

Exception flow:

- The yardstick could not be saved. The previous yardstick stays in force and the card states that the change was not applied.

TUCEW the user sees the same portfolio scored against the yardstick they chose.

### UC-13: Explore Rebalancing Scenarios



Actor: Retail Investor

Precondition: The user is authenticated and holds at least one position..

Postcondition: The user has seen the effect a hypothetical change would have on their concentration and health score. Nothing is saved and no order is placed.

TUCBW the user selects a sector in the Concentration & Rebalancing card.

Main flow:

1. The system displays the selected sector's current share of the portfolio.
2. The user requests a simulated investment into that sector.
3. The system recomputes the sector weights and the health score as they would be after that investment, and displays the before and after side by side.

Alternative flows:

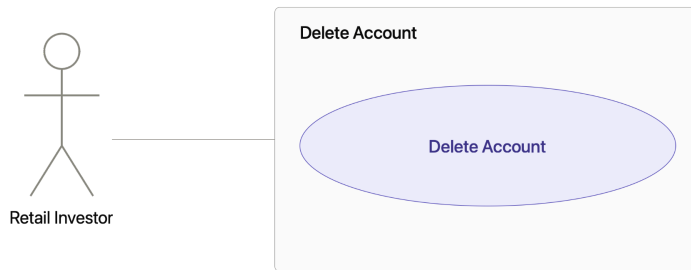
- The scenario could not be computed. The card reports this; the health score and the rest of the dashboard are unaffected.

Exception flow:

- Simulation unavailable: the scenario could not be computed. The card reports this; the health score and the rest of the dashboard are unaffected (NFR-3.1).

TUCEW the user has seen what the change would do, without having committed to it.

## UC-14: Delete Account



Actor: Retail Investor

Precondition: The user is authenticated.

Postcondition: The user's account, credentials and all associated data (portfolios, holdings, watchlist, chat conversations) have been permanently deleted and irrecoverable. The user is signed out.

TUCBW the user opens the Danger Zone on the Settings page and types their own account email address.

Main flow:

1. The system enables the Delete Account control once the typed email exactly matches the user's account email.
2. The user selects Delete Account.
3. The system displays a confirmation dialog listing exactly what will be deleted, and states that the action cannot be undone.
4. The user confirms the deletion.
5. The system deletes the user's account and all associated data from the database.
6. The system removes the user's identity from the authentication provider.
7. The system signs the user out and redirects them to the landing page.

Alternative flows:

- The typed email does not match the account's email. The Delete Account control remains disabled; nothing is deleted.
- The user cancels at the confirmation dialog. No data is deleted; the user remains signed in.

Exception flow:

- Deletion could not be fully completed where system reports that deletion and directs the user to contact support

TUCEW the user's account no longer exists and they are viewing the landing page, signed out.

## 4. Functional Requirements

Functional requirements are stated in terms of what the user can do and what the system must visibly provide.

### 4.1 Subsystem 1: Authentication

#### FR-1: User Registration

- FR-1.1: The system shall allow a new user to register with their full name, email address and password.
- FR-1.2: The system shall validate that the email address is correctly formatted before the registration is submitted.
- FR-1.3: The system shall enforce a password policy of at least 8 characters, containing at least one uppercase letter, one lowercase letter, one number and one special character.
- FR-1.4: The system shall display a clear inline error message next to each invalid or missing registration field.
- FR-1.5: The system shall reject a registration that uses an email address already associated with an existing account, and tell the user to sign in instead.

#### FR-2: Email Verification

- FR-2.1: The system shall send a 6-digit verification code to the user's email address upon registration.
- FR-2.2: The system shall require the user to enter the verification code before the account can be used.
- FR-2.3: The system shall reject invalid or expired codes with a clear error message and allow the user to retry.
- FR-2.4: The system shall direct the user to the login page once their email is verified.

#### FR-3: Login and Session

- FR-3.1: The system shall authenticate users by email address and password.
- FR-3.2: The system shall display an appropriate error message when invalid credentials are provided, without revealing whether the email or the password was incorrect.
- FR-3.3: The system shall keep the user's session active while they work, without requiring re-authentication on every page.
- FR-3.4: The system shall redirect an unauthenticated user who attempts to open a protected page to the login page.
- FR-3.5: The system shall return an authenticated user who opens the login or registration page to their dashboard.
- FR-3.6: The system shall allow the user to log out; after logging out, protected pages shall no longer be accessible until the user logs in again.

#### FR-4: Multi-Factor Authentication

- FR-4.1: The system shall require a second authentication factor, a 6-digit time-based code from an authenticator app, in addition to the password.
- FR-4.2: On first login, the system shall guide the user through pairing an authenticator app by scanning a QR code or entering a setup key manually.
- FR-4.3: The system shall allow a user to permanently delete their account and all associated portfolio data, requiring the user to confirm by typing their own email address.
- FR-4.4: On subsequent logins, the system shall prompt for the current authenticator code after the password is accepted.
- FR-4.5: The system shall reject an incorrect authenticator code with a clear error message and allow the user to retry.
- FR-4.6: Account Deletion
  - FR-4.6.1: The system shall allow an authenticated user to permanently delete their account and all associated data (portfolios, holdings, transaction history, chat conversations) upon confirmation.
  - FR-4.6.2: The system shall require the user to type their own account email address to confirm deletion before the action is enabled.
  - FR-4.6.3: The system shall display a confirmation dialog listing what will be deleted before the deletion is finalised.
  - FR-4.6.4: The system shall reject a deletion attempt where the typed email does not match the account's email, without deleting any data.
  - FR-4.6.5: The system shall sign the user out and redirect to the landing page upon successful deletion.
  - FR-4.6.6: The system shall display an error and leave the account and data intact if the deletion cannot be completed.

## 4.2 Subsystem 2: Portfolio Management

#### FR-5: File-Based Portfolio Import

- FR-5.1: The system shall accept portfolio imports in two file formats: an Excel workbook following the provided template, and a brokerage PDF statement.
- FR-5.2: The system shall allow the user to import portfolio data by uploading an Excel workbook that follows the downloadable template provided by the system.
- FR-5.3: The system shall allow the user to import portfolio data by uploading a brokerage statement, from which holdings, transactions, dividends and statement metadata are extracted automatically.
- FR-5.4: The system shall provide downloadable import templates (Excel) so the user knows the expected format.
- FR-5.5: The system shall validate every uploaded file and reject unsupported or malformed files with a clear error message that allows the user to retry.
- FR-5.6: The system shall display a preview of the parsed holdings for the user to confirm before anything is saved.

- FR-5.7: The system shall inform the user when a file parses successfully but contains no recognisable holdings, and prompt them to enter data manually or try a different file.

#### FR-6: Manual Data Management

- FR-6.1: The system shall provide a downloadable Excel template defining the expected columns, so that a user without a brokerage statement can assemble their holdings in a known format.
- FR-6.2: The system shall validate that an uploaded workbook matches the template's expected columns before parsing, and report the mismatch when it does not.

#### FR-7: Holdings Ownership and Display

- FR-7.1: The system shall associate every imported or manually entered holding with the account of the user who created it.
- FR-7.2: The system shall never display one user's portfolio data to any other user.
- FR-7.3: The system shall display the user's holdings in a structured table showing ticker symbol, company name, sector, quantity, average purchase price, current price, total value, profit/loss in Rand, and profit/loss as a percentage.
- FR-7.4: The system shall visually distinguish gains from losses (gains in green and losses in red).

### 4.3 Subsystem 3: Analytics and Financial Indicators

#### FR-8: Financial Indicator Calculation

The system shall calculate the following seven indicators from the user's actual holdings and market data. Each was selected for its value to a retail investor, as motivated in the indicator tooltips.

- FR-8.1: The system shall calculate and display the Sharpe Ratio (risk-adjusted return: portfolio return minus risk-free rate, divided by portfolio standard deviation).
- FR-8.2: The system shall calculate and display the Sortino Ratio (as the Sharpe Ratio, but penalising only downside deviation).
- FR-8.3: The system shall calculate and display the P/E Ratio for individual holdings.
- FR-8.4: The system shall calculate and display the CAPM expected return for individual holdings.
- FR-8.5: The system shall calculate and display the Altman Z-Score for individual holdings, classifying results as safe (above 2.99), grey zone (1.81 to 2.99) or distress (below 1.81).
- FR-8.6: The system shall calculate and display Beta for individual holdings, relative to a South African market proxy.
- FR-8.7: The system shall calculate and display the 14-day RSI, classifying results as overbought (above 70), neutral (30 to 70) or oversold (below 30).

- FR-8.8: The system shall calculate and display the indicator values that are within the specific preset selected.

#### FR-9: Indicator Presentation

- FR-9.1: The system shall provide a plain-language explanation for each indicator, available on hover or keyboard focus, describing what the indicator means and how to interpret its value.
- FR-9.2: The system shall display an N/A state with a short explanation when there is insufficient data to calculate an indicator, without affecting the other indicators.
- FR-9.3: The system shall colour-code each indicator value as positive, neutral or negative according to the classification thresholds defined in FR-8.
- FR-9.4: The system shall allow the user to retry a failed calculation from the affected indicator card while the rest of the page remains usable.

### 4.4 Subsystem 4: Market Data

#### FR-10: Market Data Provision

- FR-10.1: The system shall retrieve current prices, daily change percentages and trading volumes for every holding in the user's portfolio.
- FR-10.2: The system shall retrieve historical price data over selectable periods for charting and indicator calculation.
- FR-10.3: The system shall allow the user to search for a stock by ticker symbol or company name and see matching results.
- FR-10.4: The system shall continue functioning when live market data for a holding is unavailable, showing the most recent known values or a clear unavailable state for that holding only.

### 4.5 Subsystem 5: News Correlation

#### FR-11: Price Movement Detection

- FR-11.1: The system shall monitor daily price movements for every holding in the user's portfolio.
- FR-11.2: The system shall flag a movement as significant when the day's price change exceeds a configured threshold (default: 5% in either direction).

#### FR-12: News Retrieval and Correlation

- FR-12.1: The system shall retrieve financial news articles from its news data providers published within a 48-hour window of a detected significant price movement.
- FR-12.2: The system shall show only articles relevant to the specific holding's ticker symbol or company name.
- FR-12.3: The system shall display each correlated article with its headline, source, publication time, related ticker badge, and an indicator of how the related stock's price moved.

- FR-12.4: The system shall allow the user to filter displayed articles by ticker symbol.
- FR-12.5: The system shall allow the user to filter displayed articles by date range.
- FR-12.6: The system shall clearly indicate when no news is available for a holding or filter combination, and prompt the user to broaden their filters.

## 4.6 Subsystem 6: AI Assistant

### FR-13: Conversational Interface

- FR-13.1: The system shall provide a chat interface where the user can submit natural-language questions about their portfolio.
- FR-13.2: The system shall ground every response in the user's own data by combining their holdings, calculated indicators and relevant market data with the question before it reaches the AI model.
- FR-13.3: The system shall display suggested prompts when the chat is empty, which the user can send with a single click.
- FR-13.4: The system shall display an error message and allow retry when a response cannot be generated.

### FR-14: Conversation Management

- FR-14.1: The system shall save the user's conversations so they persist across sessions.
- FR-14.2: The system shall display the user's saved conversations in a list, and load a conversation's full message history when selected.
- FR-14.3: The system shall allow the user to start a new conversation at any time.
- FR-14.4: The system shall allow the user to rename a saved conversation.
- FR-14.5: The system shall allow the user to delete a saved conversation.

### FR-15: Compliance and Guardrails

- FR-15.1: The system shall never provide explicit buy or sell recommendations, regardless of how a question is phrased.
- FR-15.2: When the user asks for explicit investment advice, the system shall respond with a disclaimer explaining that EquityLens provides analysis and information only, and direct the user to the Analytics page.
- FR-15.3: The system shall display a disclaimer with AI responses clarifying that they are informational and not regulated financial advice.

## 4.7 Subsystem 7: Dashboard and Insights

### FR-16: Portfolio Overview

- FR-16.1: The system shall display the portfolio's total value, total gain/loss (Rand and percentage), and today's change.
- FR-16.2: The system shall summarise, in one plain-language sentence, how the portfolio moved today and which holding drove the move.

- FR-16.3: The system shall display the user's largest positions with each position's share of the total portfolio and its daily change.
- FR-16.4: The system shall display the portfolio's sector allocation, and allow the user to select a sector to see the holdings within it.

#### FR-17: Performance vs Benchmark

- FR-17.1: The system shall record one snapshot per day of the portfolio's total value, and shall derive a comparison benchmark weighted to match the portfolio's own regional exposure (South African, US and global equity), converting foreign components at the prevailing USD/ZAR rate.
- FR-17.2: The system shall chart portfolio value and benchmark together over time, and display the portfolio return, benchmark return, and the difference for the charted window.
- FR-17.3: The system shall leave a visible gap in the benchmark series for days on which the benchmark could not be fetched, rather than substituting a fabricated value.
- FR-17.4: The system shall display monthly dividend income derived from the user's imported dividend records.

#### FR-18: Portfolio Health and Action Centre

- FR-18.1: The system shall compute a portfolio health score out of 10 from the user's actual holdings, as a weighted combination of three structural risk factors: sector concentration, single-position concentration, and effective portfolio breadth.
- FR-18.2: The system shall display the health score with a classification (Excellent, Healthy, Mixed, Needs attention) and identify the strongest and weakest contributing factors in plain language.
- FR-18.3: The system shall flag an actionable item when a defined risk threshold is crossed: a single sector at 40% or more of the portfolio, a single holding at 30% or more, or fewer than 5 positions held.
- FR-18.4: Each actionable item shall state the risk in plain language and link the user to the page or section where they can act on it.
- FR-18.5: The system shall show a reassuring all-clear state when no thresholds are crossed, and an onboarding prompt when the user has no holdings.

#### FR-19: General Interface

- FR-19.1: The system shall allow the user to switch between light and dark display themes, and remember the choice.
- FR-19.2: The system shall display sector-level market context derived from the price movements of the user's own holdings, labelled as such so it is not mistaken for an external market feed.
- FR-19.3: The system shall provide an in-product Help menu giving the user access to guidance, tutorials and frequently asked questions.

## 4.8 Subsystem 8: Watchlist

### FR-20: Watchlist Management

- FR-20.1: The system shall allow the user to add a stock they do not own to their watchlist by searching for it by ticker or company name.
- FR-20.2: The system shall display each watchlist entry with its ticker, company name, sector, current price and daily percentage change.
- FR-20.3: The system shall allow the user to remove an entry from their watchlist.
- FR-20.4: The system shall keep each user's watchlist private to that user.

### FR-21: Health Yardstick Configuration

- FR-21.1: The system shall provide a set of named scoring strategies, each defining the weights and thresholds the health score is measured against, with a plain-language description of what it optimises for.
- FR-21.2: The system shall allow the user to select a strategy, and shall recompute and redisplay the health score against it without altering the underlying holdings.
- FR-21.3: The system shall persist the user's selected strategy across sessions.
- FR-21.4: In the absence of an explicit selection, the system shall apply the strategy matched to the user's stated goal horizon, and shall allow the user to return to it.
- FR-21.5: The system shall make clear that changing the strategy changes what the portfolio is compared to, not the risk the portfolio carries.

### FR-22: Rebalancing Simulation

- FR-22.1: The system shall allow the user to simulate an additional investment into a chosen sector and display the resulting sector weights and health score alongside the current ones.
- FR-22.2: The system shall allow the user to simulate levelling the allocation across all sectors and display the resulting health score.
- FR-22.3: The system shall not save any simulated change and shall not place any order; simulations are illustrative only.

## 5. Non-Functional Requirements

### 5.1 NFR-1: Security

- NFR-1.1: 100% of API endpoints, other than the health check and the registration/login/verification endpoints, shall reject unauthenticated requests with an HTTP 401 response.
- NFR-1.2: User credentials shall never be stored in plain text by the system. Sessions shall be established with expiring access tokens with a lifetime of at most 60 minutes; an expired session shall require re-authentication or a silent token refresh before any protected resource is served.
- NFR-1.3: Multi-factor authentication (TOTP) shall be enforced on user accounts.
- NFR-1.4: All data in transit shall be encrypted with TLS; all data at rest shall be encrypted with AES-256.
- NFR-1.5: All portfolio data shall be scoped to the owning user at the data-access level: 0 instances of cross-user data exposure, verified by integration tests.
- NFR-1.6: The system shall never store raw brokerage credentials; portfolio data enters the system only by file upload or manual entry.
- NFR-1.7: All personal data handling shall comply with POPIA.

### 5.2 NFR-2: Performance

- NFR-2.1: API endpoints shall respond within 500 ms at the 85th percentile under normal load.
- NFR-2.2: Initial page load shall complete within 5 seconds on a standard broadband connection.
- NFR-2.3: Dashboard visualisations shall render within 3 seconds for portfolios of up to 50 holdings.
- NFR-2.4: Market data shall be served through a cache-aside store: a request for an instrument that is already cached shall be answered with zero calls to the upstream provider, keeping repeat dashboard loads within provider rate limits.

### 5.3 NFR-3: Reliability

- NFR-3.1: The failure of any single external data source (market data, news, AI model) shall degrade only its own feature; the remainder of the system shall stay fully functional.
- NFR-3.2: If the primary AI service fails, the fallback shall activate within 2 seconds with no user-visible error.
- NFR-3.3: A failed fetch for one holding's market data shall not prevent the dashboard from rendering the remaining holdings.

### 5.4 NFR-4: Scalability

- NFR-4.1: The backend shall support at least 50 concurrent users without performance degradation beyond the NFR-2 targets.
- NFR-4.2: Database query times shall remain below 200 ms at the 95th percentile with up to 1 million rows of historical market data.

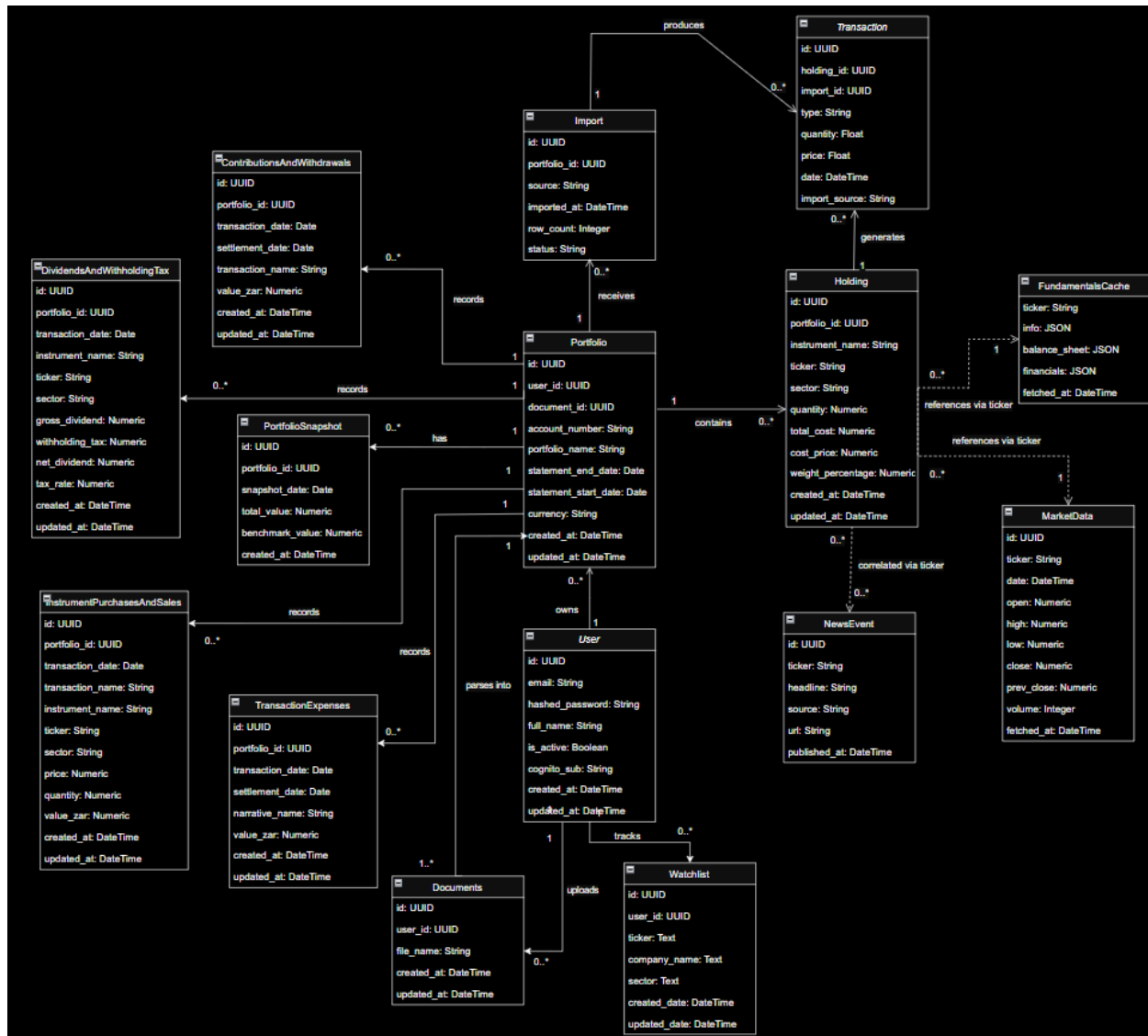
- NFR-4.3: The system shall accommodate a 200% workload increase without major architectural changes and with no more than a 10% performance decrease.

## 5.5 NFR-5: Usability and Accessibility

- NFR-5.1: A new user shall be able to register, import a portfolio and reach a populated dashboard within 5 minutes of first use.
- NFR-5.2: The interface shall meet WCAG 2.1 AA: all interactive elements keyboard-navigable with visible focus states, and text contrast ratios of at least 4.5:1.
- NFR-5.3: Every financial indicator shall have a plain-language explanation accessible from where the indicator is displayed, so no finance training is required to use the system.
- NFR-5.4: The interface shall be fully responsive across desktop (1280px+), tablet (768-1279px) and mobile (320-767px) with no horizontal scrolling on mobile.
- NFR-5.5: The interface shall respect the user's reduced-motion preference and disable non-essential animations when set.

## 6. Domain Model

### 6.1 UML Class Diagram



### 6.2 Entity Descriptions

**User** - an account holder. `cognito_sub` links the local record to the identity provider; no password material is stored locally for identity-provider accounts. Each user's data (documents, portfolios, watchlist, conversations) is reachable only through their own account.

**Document** - a record of an uploaded statement file (PDF/Excel). Retained so an import can be traced back to its source file.

**Portfolio** - a set of holdings imported from one statement or created manually, including the statement period it covers. A user may hold multiple portfolios.

**Holding** - a single position within a portfolio: the instrument, quantity held, and cost basis (total cost and per-share cost price). Current value is derived at read time from market data, not stored.

**InstrumentTransaction** - a buy or sell recorded in the statement, with date, price, quantity and Rand value.

**ContributionWithdrawal** - cash paid into or withdrawn from the account.

**DividendRecord** - a dividend event including gross amount, withholding tax and net amount, used for the dividend income view.

**TransactionExpense** - fees and costs recorded in the statement.

**PortfolioSnapshot** - one row per portfolio per day recording total portfolio value and the JSE All Share Index level, accumulated to build the performance-vs-benchmark history. Uniqueness is enforced on (portfolio, date).

**WatchlistEntry** - a stock the user tracks without owning. Private to the user.

**Conversation** - a saved AI assistant conversation with a user-editable title.

**Message** - a single message within a conversation; role distinguishes the user's messages from the assistant's.

**MarketData (cache)** - fetched price data (price, volume, daily change, fetch timestamp) retained to serve repeat requests within the cache window. It is a cache, not a system of record.

**NewsEvent (cache)** - a retrieved news article (headline, source, URL, publication time, related ticker) retained for display and correlation

## 7. Requirements-Use Case Traceability

|       | UC-1 | UC-2 | UC-3 | UC-4 | UC-5 | UC-6 | UC-7 | UC-8 | UC-9 | UC-10 | UC-11 | UC-12 | UC-13 |
|-------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|
| FR-1  |      |      |      |      |      | X    |      |      |      |       |       |       |       |
| FR-2  |      |      |      |      |      | X    |      |      |      |       |       |       |       |
| FR-3  | X    | X    | X    | X    | X    |      | X    | X    | X    | X     | X     | X     | X     |
| FR-4  |      |      |      |      |      |      | X    |      |      |       |       |       |       |
| FR-5  | X    |      |      |      |      |      |      |      |      |       |       |       |       |
| FR-6  | X    |      |      |      |      |      |      |      |      |       |       |       |       |
| FR-7  | X    | X    |      |      |      |      |      |      |      |       |       |       |       |
| FR-8  |      |      | X    | X    |      |      |      |      |      |       |       |       |       |
| FR-9  |      |      | X    |      |      |      |      |      |      |       |       |       |       |
| FR-10 |      | X    | X    |      | X    |      |      | X    |      | X     |       |       |       |
| FR-11 |      |      |      |      | X    |      |      |      |      |       |       |       |       |

[illegible]

## 8. Assumptions and Dependencies

### 8.1 Assumptions

- A-1: Users have a basic understanding of stock ticker symbols and financial terminology.
- A-2: Users have an existing investment portfolio with at least one holding, or intend to build one.
- A-3: Users have a modern web browser with JavaScript enabled, and access to an authenticator app on a phone or desktop for MFA.
- A-4: Portfolio PDF statements are sourced from supported South African brokerages, or follow the provided import templates.
- A-5: Market data supplied by the market data provider accurately reflects JSE-listed stock prices.

### 8.2 Dependencies

- D-1: Primary market data provider. If unavailable: the fallback provider activates; if both fail, the last cached values are shown.
- D-2: News data providers. If unavailable: the news correlation feature is unavailable; the rest of the system is unaffected.
- D-3: AI model provider. If unavailable: the assistant falls back to the secondary layer; if both fail, an error prompts the user to retry.
- D-4: Cloud free-tier allowance. If exceeded: infrastructure costs would be incurred, violating the project's cost constraint.
- D-5: CI pipeline provider. If unavailable: automated quality gates are unavailable and manual testing is required before merges.
- D-6: PDF text extraction capability. If unavailable: PDF import is unavailable; Excel and manual entry remain functional.

# **Non-functional requirement (NFR) Testing**

To confirm that the non-functional requirements specified in the SRS are met, our team (EquityLens) will employ both executable and repeatable tests. We will compare the actual outcome to the objective specified in our SRS document when each quality requirement is met utilizing a method or tool. Additionally, we have included a screenshot and test outputs that serve as evidence of whether each of the following requirements has been met.

## **How did we do the testing?**

Every NFR has its own tool, such as OWSAP or K6, and we conducted all testing using the GitHub action workflow. Once it has been pushed to either dev or main, the outcome may be viewed to determine whether it was successful or unsuccessful. Next, we took the screenshot that is displayed below. Some other test were run locally and screenshot were taken as evidence

## Security Testing

| ID             | Executable tests/<br>procedures                                                                                                             | Tool                           | Target/ expected results                                                                                      | Actual result/<br>evidence |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------|
| <b>NFR-1.1</b> | Attempting to use every protected API without logging in.                                                                                   | OWSAP ZAP                      | 100 % of protected routes return HTTP 401.                                                                    | <b>See evidence below</b>  |
| <b>NFR-1.2</b> | To check that all the passwords are hashed in the database, tokens expire within 60 minutes, and of course all expired tokens are rejected. | Live system test + AWS Cognito | The database will not have a plain text password; tokens that are older than sixty minutes will be discarded. | <b>See evidence below</b>  |
| <b>NFR-1.3</b> | To try to log in with an invalid TOTP, then with a valid TOTP.                                                                              | Live Login Test                | The login will only be succeeded, if the TOTP is valid.                                                       | <b>See evidence below</b>  |
| <b>NFR-1.4</b> | To check the websites use “HTTPS” and not “HTTP” and all the stored data is encrypted.                                                      | OpenSSL                        | The HTTP/TLS is used on the website.                                                                          | <b>See evidence below</b>  |
| <b>NFR-1.5</b> | All portfolio data shall be scoped to the owning user at the data-access level: 0 instances of cross-user data exposure.                    | OWSAP ZAP                      | No one should be able to access other people portfolio                                                        | <b>See evidence below</b>  |

|                |                                                                                               |           |                                               |                           |
|----------------|-----------------------------------------------------------------------------------------------|-----------|-----------------------------------------------|---------------------------|
| <b>NFR-1.6</b> | Looks for any brokerage usernames, passwords, and API keys in the database.                   | GitLeaks  | There are 0 raw brokerage credentials stored. | <b>See evidence below</b> |
| <b>NFR-1.7</b> | This checks that the website (EquityLens) follows all the rules of POPIA privacy requirement. | OWSAP ZAP | All the defined POPIA acceptance passes.      | <b>See evidence below</b> |

## Evidence of the Security Testing

### NFR-1.1 and 1.7 and 1.5

In here we run the OWSAP command locally in order to test it, and we only got warning errors

```
PS C:\Users\Abdul\Documents\Equity-Lens> docker run --rm -t `
>> --network equity-lens_default `
>> ghcr.io/zaproxy/zaproxy:stable `
>> zap-api-scan.py `
>> -t http://equity-lens-backend-1:8000/openapi.json ` ...
WARN-NEW: X-Content-Type-Options Header Missing [10021] x 3
    http://equity-lens-backend-1:8000/openapi.json (200 OK)
    http://equity-lens-backend-1:8000/health (200 OK)
    http://equity-lens-backend-1:8000/api/news/test-aapl/ticker (200 OK)
WARN-NEW: Cross-Origin-Resource-Policy Header Missing or Invalid [90004] x 3
    http://equity-lens-backend-1:8000/openapi.json (200 OK)
    http://equity-lens-backend-1:8000/health (200 OK)
    http://equity-lens-backend-1:8000/api/news/test-aapl/ticker (200 OK)
WARN-NEW: Application Error Disclosure [90022] x 1
    http://equity-lens-backend-1:8000/api/auth/mfa/verify-login (500 Internal Server Error)
FAIL-NEW: 0    FAIL-INPROG: 0    WARN-NEW: 4    WARN-INPROG: 0    INFO: 0    IGNORE: 0    PASS: 115
```

## NFR-1.2

### 1). Password is hashed

**Users** [Info](#)

View, edit, and create users in your user pool. Users that are enabled and confirmed can sign in to your user pool.

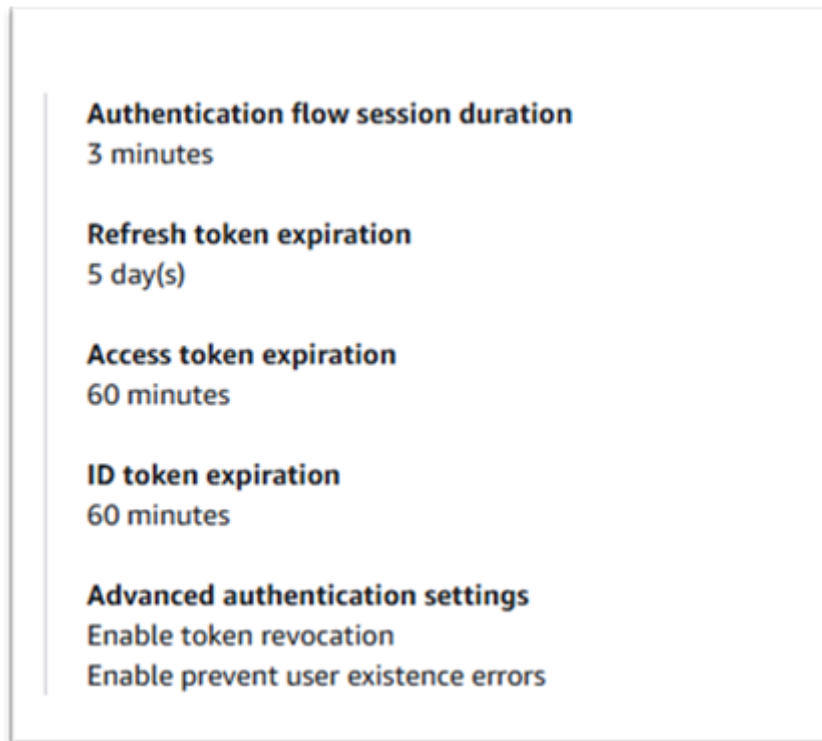
Property: User name

[Refresh](#) [Delete user](#) [Create user](#)

|                       | User name                                            | Email address            | Email verified | Confirmation status | Status  |
|-----------------------|------------------------------------------------------|--------------------------|----------------|---------------------|---------|
| <input type="radio"/> | <a href="#">914c3248-e091-708d-ca34-69df14bf...</a>  | zaprox@example.com       | No             | Unconfirmed         | Enabled |
| <input type="radio"/> | <a href="#">41bc22b8-c051-70d9-343e-8e9f4eff6...</a> | joshuaheath004@gmail.com | Yes            | Confirmed           | Enabled |
| <input type="radio"/> | <a href="#">c13c2288-a081-7088-b847-99a1d2b3...</a>  | wedsaq5@gmail.com        | Yes            | Confirmed           | Enabled |
| <input type="radio"/> | <a href="#">e13c32b8-c061-70fd-bc3c-e1440e38f...</a> | 123@tuks.co.za           | No             | Unconfirmed         | Enabled |
| <input type="radio"/> | <a href="#">111c3278-2061-7010-9407-a8f9b72a...</a>  | floshheath@gmail.com     | Yes            | Confirmed           | Enabled |
| <input type="radio"/> | <a href="#">d1ace2e8-a071-702c-21b4-d301008f...</a>  | thefivebigtb5@gmail.com  | No             | Unconfirmed         | Enabled |
| <input type="radio"/> | <a href="#">815ce248-3001-70ff-8c89-747f2bb07...</a> | test@example.com         | No             | Unconfirmed         | Enabled |
| <input type="radio"/> | <a href="#">41dc1268-d0c1-70d8-163b-d82b982...</a>   | officialiceyxd@gmail.com | Yes            | Confirmed           | Enabled |
| <input type="radio"/> | <a href="#">715c0278-e081-70bf-ba29-8c064fe69...</a> | u24590739@tuks.co.za     | Yes            | Confirmed           | Enabled |
| <input type="radio"/> | <a href="#">916c42c8-a0f1-7055-6f18-0b6cc39a4...</a> | floshheath+t1@gmail.com  | No             | Unconfirmed         | Enabled |

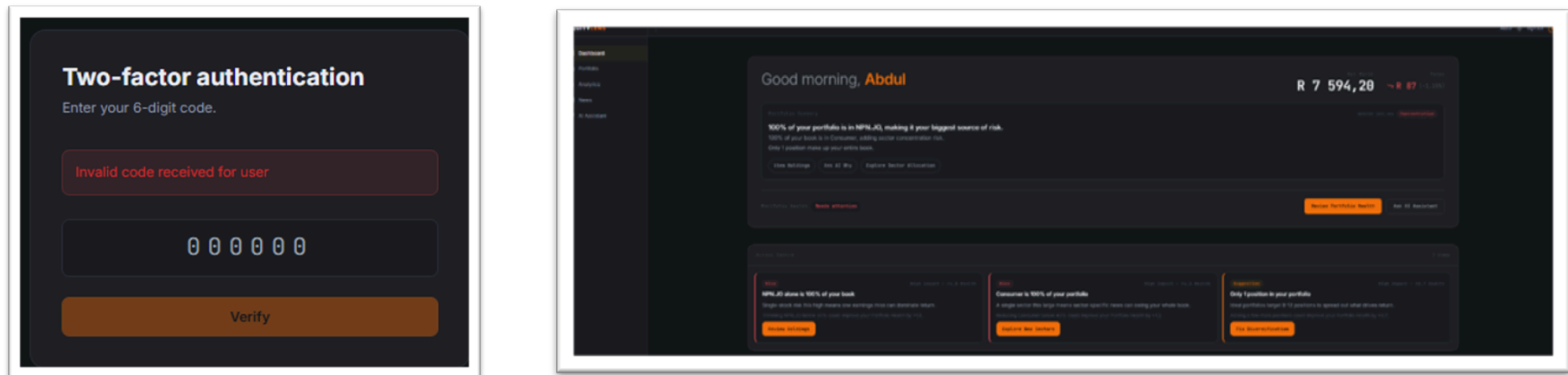
So, the database does not store the password, and the password is protected by Cognito which does not display the passwords in the AWS console for security reasons.

## 2). Token expiry date



As we can see from the screenshot that both the access token and the ID token are configured in AWS to expire after 30 minutes

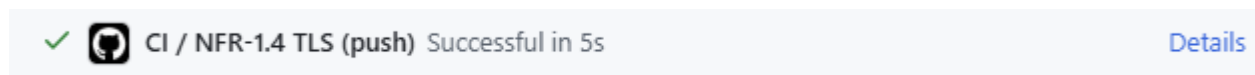
### NFR-1.3



After logging in, he must verify his password using Google Authenticator. If it is incorrect, an error notice appears on the left, and if it is valid, I will be taken to the dashboard, which is visible on the right.

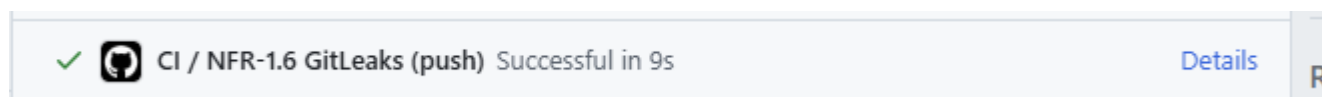
### NFR 1.4

This test was conducted in the pipelines, and when it passed, a screenshot is displayed below.



### NFR 1.6

This test was conducted in the pipelines, and when it passed, a screenshot is displayed below.



## Performance Testing

| ID      | Executable tests/<br>procedures                                                                               | Tool          | Target/ expected results                                                           | Actual result/ evidence |
|---------|---------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------------------------------------------|-------------------------|
| NFR-2.1 | sending the API request and timing the response.                                                              | K6            | 85% of requests will be answered in 500 milliseconds.                              | See evidence below      |
| NFR-2.2 | Measure how long it takes for the first page to load after opening the website (live).                        | Lighthouse    | The page needs to load in 10 seconds or less.                                      | See evidence below      |
| NFR-2.3 | To measure how long it takes for the charts and graphs to show after opening the dashboard page.              | Lighthouse    | All the holdings graphs and charts (dashboard visualizations) appear in 6 seconds. | See evidence below      |
| NFR-2.4 | To make two requests for the identical market data and verify that the second request makes use of the cache. | Live API Test | The second call makes 0 calls to the external provider.                            | See evidence below      |

# Evidence of the Performance Testing

## NFR-2.1

```
PS C:\Users\Abdul\Documents\Equity-Lens> k6 run backend/tests/K6/performance.js

THRESHOLDS

http_req_duration
✓ 'p(85)<500' p(85)=75.04ms

http_req_failed
✓ 'rate<0.01' rate=0.00%

TOTAL RESULTS

checks_total.....: 25312 840.237302/s
checks_succeeded...: 100.00% 25312 out of 25312
checks_failed.....: 0.00% 0 out of 25312

✓ status is 200

HTTP
http_req_duration.....: avg=57.04ms min=22.3ms med=49.75ms max=590.72ms p(90)=84ms p(95)=98.14ms
{ expected_response:true }...: avg=57.04ms min=22.3ms med=49.75ms max=590.72ms p(90)=84ms p(95)=98.14ms
http_req_failed.....: 0.00% 0 out of 25312
http_reqs.....: 25312 840.237302/s

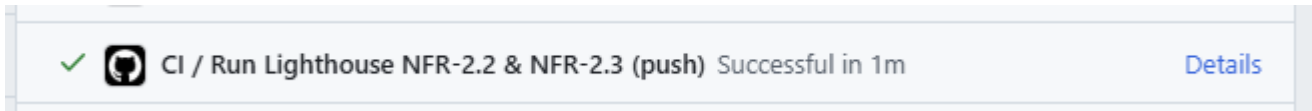
EXECUTION
iteration_duration.....: avg=59.29ms min=24.32ms med=50.04ms max=2.09s p(90)=86.22ms p(95)=101.42ms
iterations.....: 25312 840.237302/s
vus.....: 50 min=50 max=50
vus_max.....: 50 min=50 max=50

NETWORK
data_received.....: 6.4 MB 213 kB/s
data_sent.....: 3.1 MB 103 kB/s

running (0m30.1s), 00/50 VUs, 25312 complete and 0 interrupted iterations
default ✓ [=====] 50 VUs 30s
```

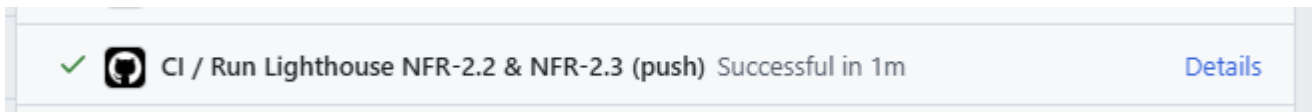
## NFR-2.2

This test was conducted in the pipelines, and when it passed, a screenshot is displayed below.



## NFR-2.3

This test was conducted in the pipelines, and when it passed, a screenshot is displayed below.



## NFR-2.4

```
Local price hit: CSP500.JO  
Local price hit: CTOP50.JO  
Local price hit: EASYAI.JO  
Local price hit: STXEMG.JO
```

```
Fundamentals cache hit: MTN  
Fundamentals cache hit: NPN.JO  
Fundamentals cache hit: CSP500.JO  
Fundamentals cache hit: CTOP50.JO  
Fundamentals cache hit: EASYAI.JO  
Fundamentals cache hit: STXEMG.JO
```

So, here are the backend logs, the one on the left is shown when the user clicked the analytics page for the first time and the second image on the left is when the user presses the analytics page (for the second time) it say's "cache hit" in the backend logs.

## Reliability Testing

These tests were run through the pipelines(testing) and this screenshot taken from there.

| ID      | Executable tests/ procedures                                                                   | Tool   | Target/ expected results                                                               | Actual result/ evidence |
|---------|------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------------------------------------|-------------------------|
| NFR-3.1 | Make the Market, News, or AI service fail, then see if the other aspects are still functional. | Pytest | Except for the feature that failed, the remaining features should still be functional. | See evidence below      |
| NFR-3.2 | Make the AI fail and see if the API can manage it.                                             | Pytest | The API does not crash the system and provides an accurate response.                   | See evidence below      |
| NFR-3.3 | Make the market data fail for one or more stocks in the system.                                | Pytest | The other stocks should still appear without the failing one.                          | See evidence below      |

## Evidence Reliability Testing

### NFR-3.1

```
33 tests/integration/test_nfr_failure_isolation.py . [ 6%]
```

### NFR-3.2

```
32 tests/integration/test_nfr_ai_failure.py . [ 5%]
```

### NFR-3.3

```
34 tests/integration/test_nfr_holding_failure.py . [ 6%]
```

## Scalability Testing

All these tests were done locally, and the results are shown below with a photo

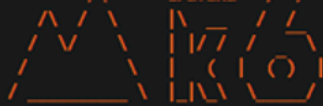
| ID      | Executable tests/<br>procedures                                      | Tool | Target/ expected results                                                                                            | Actual result/<br>evidence |
|---------|----------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------------------|----------------------------|
| NFR-4.1 | Test the system simultaneously with 50 users.                        | K6   | Here, we must ensure that 99% of all requests are successful and that 95% of them respond in less than two seconds. | See evidence below         |
| NFR-4.2 | Send 100 simultaneous API Requests to EquityLens                     | K6   | At least 99% of requests succeed                                                                                    | See evidence below         |
| NFR-4.3 | To compare their performances while increasing the workload by 200%. | K6   | little significant system modifications are required, and performance drops by little more than 10%.                | See evidence below         |



# Evidence Scalability Testing

## NFR-4.1

```
PS C:\Users\Abdul\Documents\Equity-Lens> k6 run backend/tests/K6/scalability_nfr_4.1.js
```



```
execution: local
script: backend/tests/K6/scalability_nfr_4.1.js
output: -
```

```
scenarios: (100.00%) 1 scenario, 50 max VUs, 1m0s max duration (incl. graceful stop):
  * default: 50 looping VUs for 30s (gracefulStop: 30s)
```

### THRESHOLDS

```
checks
✓ 'rate==1.0' rate=100.00%

http_req_duration
✓ 'p(95)<2000' p(95)=233.15ms

http_req_failed
✓ 'rate<0.01' rate=0.00%
```

### TOTAL RESULTS

```
checks_total.....: 14954 480.694174/s
checks_succeeded...: 100.00% 14954 out of 14954
checks_failed.....: 0.00% 0 out of 14954
```

```
✓ status is 200
```

```
HTTP
http_req_duration.....: avg=99ms min=24.1ms med=80.58ms max=725.45ms p(90)=162.69ms p(95)=233.15ms
{ expected_response:true }...: avg=99ms min=24.1ms med=80.58ms max=725.45ms p(90)=162.69ms p(95)=233.15ms
http_req_failed.....: 0.00% 0 out of 14954
http_reqs.....: 14954 480.694174/s
```

```
EXECUTION
iteration duration.....: avg=101.71ms min=24.1ms med=81.29ms max=1.27s p(90)=166.72ms p(95)=237.04ms
iterations.....: 14954 480.694174/s
vus.....: 3 min=3 max=50
vus_max.....: 50 min=50 max=50
```

```
NETWORK
data_received.....: 3.9 MB 124 kB/s
data_sent.....: 1.8 MB 59 kB/s
```

```
running (0m31.1s), 00/50 VUs, 14954 complete and 0 interrupted iterations
```

## NFR-4.2

```
PS C:\Users\Abdul\Documents\Equity-Lens> k6 run backend/tests/K6/scalability_nfr_4.2.js
```



```
execution: local
script: backend/tests/K6/scalability_nfr_4.2.js
output: -
```

```
scenarios: (100.00%) 1 scenario, 50 max VUs, 1m0s max duration (incl. graceful stop):
    * default: 50 looping VUs for 30s (gracefulStop: 30s)
```

### THRESHOLDS

```
checks
✓ 'rate==1.0' rate=100.00%
```

### TOTAL RESULTS

```
checks_total.....: 9689      318.260783/s
checks_succeeded...: 100.00% 9689 out of 9689
checks_failed.....: 0.00%    0 out of 9689
```

```
✓ status is 200
```

### HTTP

```
http_req_duration.....: avg=152.85ms min=39.89ms med=126.57ms max=766.68ms p(90)=268.01ms p(95)=365.72ms
    { expected_response:true }...: avg=152.85ms min=39.89ms med=126.57ms max=766.68ms p(90)=268.01ms p(95)=365.72ms
http_req_failed.....: 0.00%    0 out of 9689
http_reqs.....: 9689      318.260783/s
```

### EXECUTION

```
iteration_duration.....: avg=155.11ms min=39.89ms med=127.33ms max=773.09ms p(90)=275.77ms p(95)=367.77ms
iterations.....: 9689      318.260783/s
vus.....: 50      min=50      max=50
vus_max.....: 50      min=50      max=50
```

### NETWORK

```
data_received.....: 2.5 MB 82 kB/s
data_sent.....: 1.2 MB 39 kB/s
```

```
running (0m30.4s), 00/50 VUs, 9689 complete and 0 interrupted iterations
default ✓ [=====] 50 VUs 30s
```

### NFR-4.3

```
PS C:\Users\Abdul\Documents\Equity-Lens> k6 run backend/tests/K6/scalability_nfr_4.3.js
```

```
execution: local
```

```
script: backend/tests/K6/scalability_nfr_4.3.js
```

output: -

```
scenarios: (100.00%) 2 scenarios, 200 max VUs, 1m35s max duration (incl. graceful stop):
```

```
* baseline: 50 looping VUs for 30s (gracefulStop: 30s)
```

```
* highload: 150 looping VUs for 30s (startTime: 35s, gracefulStop: 30s)
```

## THRESHOLDS

```
http_req_duration{load:baseline}
```

✓ 'p(85)<500' p(85)=251.7ms

```
http_req_duration{load:high}
```

✓  $p(85) < 550$   $p(85) = 246.02 \text{ ms}$

http\_req\_failed

✓ 'rate<0.01' rate=0.00%

## TOTAL RESULTS

```
checks_total.....: 19915    209.612571/s
```

```
checks_succeeded...: 100.00% 19915 out of 19915
```

```
checks failed.....: 0.00% 0 out of 19915
```

✓ status is 200

HTTP

```
http_req_duration.....: avg=272.63ms min=28.91ms med=136.64ms max=17.98s p(90)=337.98ms p(95)=428.11ms
```

```
{ expected_response:true }...: avg=272.63ms min=28.91ms med=136.64ms max=17.98s p(90)=337.98ms p(95)=428.11ms
```

```
{ load:baseline }.....: avg=179.18ms min=28.91ms med=133.61ms max=1.82s p(90)=324.52ms p(95)=420.08ms
```

```
{ load:high }.....: avg=336.41ms min=48.89ms med=138.29ms max=17.98s p(90)=347.16ms p(95)=466.85ms
```

```
http_req_failed.....: 0.00% 0 out of 19915
```

```
http_reqs.....: 19915 209.612571/s
```

## EXECUTION

```
iteration_duration.....: avg=283.8ms min=28.91ms med=137.08ms max=17.98s p(90)=350.92ms p(95)=498.86ms
```

```
iterations.....: 19915    209.612571/s
```

```

vus max.....: 200      min=200      max=200

```

## NETWORK

```
data received.....: 5.2 MB 55 kB/s
```

```
data_sent.....: 2.6 MB 28 kB/s
```

```
running (1m35.0s), 000/200 VUs, 19915 complete and 135 interrupted iterations
```

```

baseline ✓ [=====] 50 Ws  30s

```

```
highload ✓ [=====] 150 VUs 30s
```

C:\Users\Abdul\Documents\Equity Loans

## Usability and Accessibility Testing

Some of these tests were done locally

| ID             | Executable tests/ procedures                                                                   | Tool                                | Target/ expected results                                                                             | Actual result/ evidence   |
|----------------|------------------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------|
| <b>NFR-5.1</b> | We measure the time while the user registers, logs in, imports a PDF, and opens the dashboard. | Live system Testing                 | Complete everything within 5 minutes                                                                 | <b>See evidence below</b> |
| <b>NFR-5.2</b> | In here we need to check the keyboard navigation and the focus and the text contrast           | Live system tests + chrome DevTools | The keyboard works; the focus must be visible + constant ratio > 4.5:1                               | <b>See evidence below</b> |
| <b>NFR-5.3</b> | Every financial indicator in the system should have a simple explanation for the user.         | Live system tests                   | Every indicator has a simple explanation                                                             | <b>See evidence below</b> |
| <b>NFR-5.4</b> | The website must be accessible on a desktop, laptop, and phone.                                | Playwright                          | A website that doesn't require horizontal scrolling can function on many screens and mobile devices. | <b>See evidence below</b> |

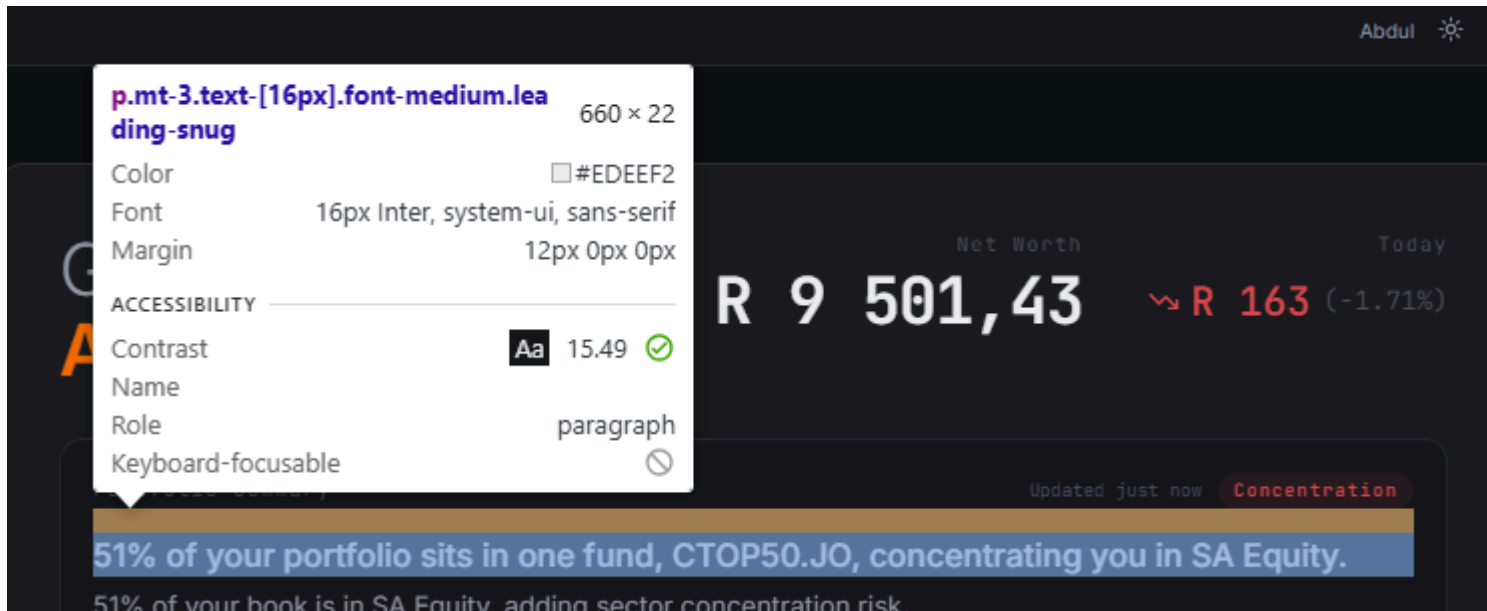
# Evidence Usability and Accessibility Testing

NFR-5.1



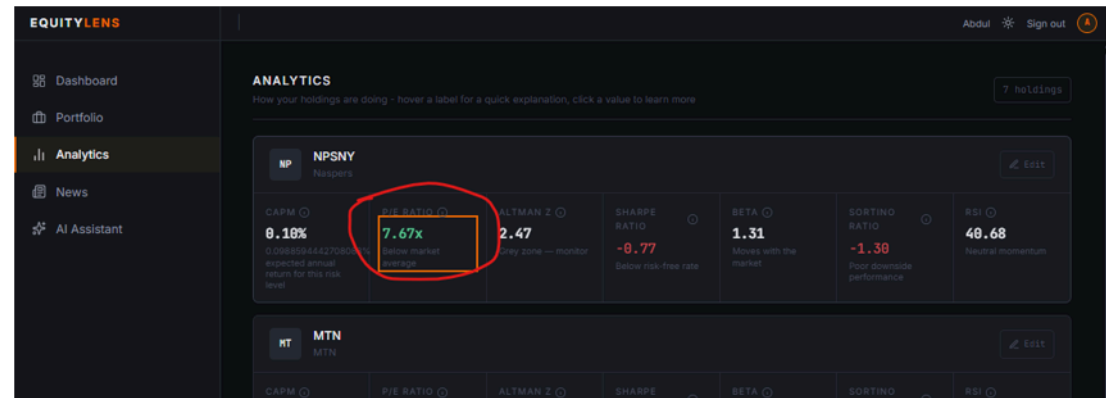
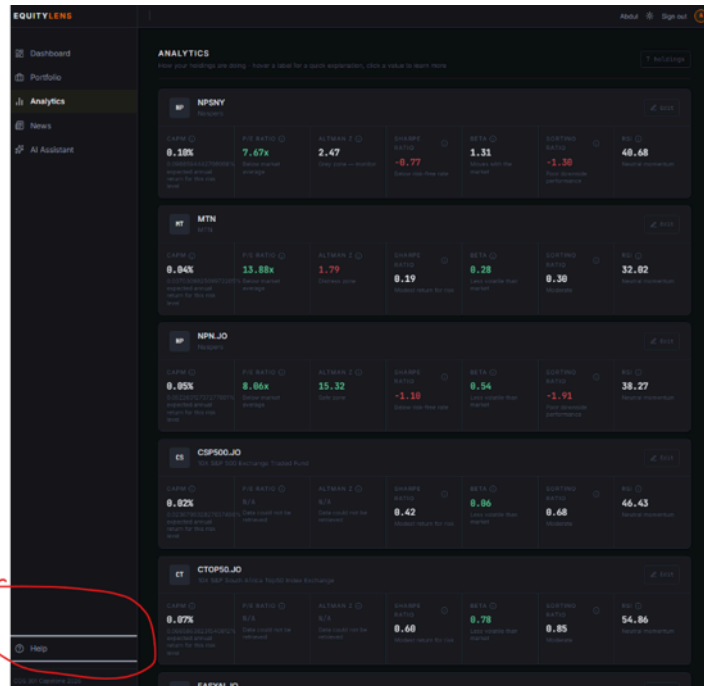
So, in here we did a timer and tested how long it takes for the user to register, then log in, then import the pdf, then view the dashboard as seen above from the timer, it was less than 5 minutes and there was still 1 minutes and 28 seconds left.

## NFR-5.2

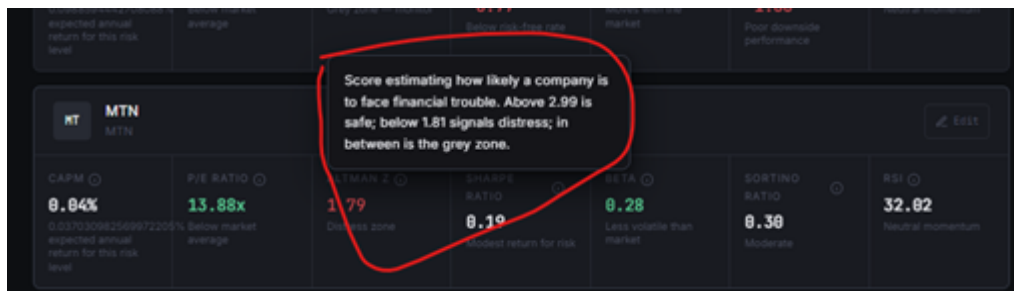
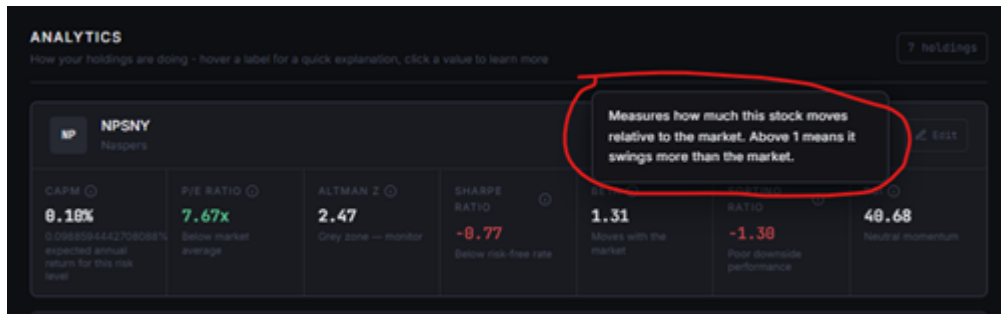


In here we can see that it meets our requirements (constant ratio > 4.5:1) - for the above image

In here we can see that when pressing the tab, it moves from one place to another, this meets our requirement for 5.2



### NFR-5.3



In the images above, we can see that if you don't understand any of this calculation you can hover on any of them, then an explanation will be given.

## NFR-5.4

This test was run locally and as seen below it has passed

```
PS C:\Users\Abdul\Documents\Equity-Lens\frontend> npx playwright test test/nfr/nfr_5_4.spec.js

Running 32 tests using 1 worker
  32 passed (2.9m)

To open last HTML report run:

  npx playwright show-report
```

# Executable and Repeatable NFR Tests

| ID      | Quantified requirement                                                                                                                                                                                                                                                                     | Tactics in SAS                                                                                     | Test/Tool                      | Target/actual                                                     |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------|
| NFR-1.1 | 100% of API endpoints, other than the health check and the registration/login/verification endpoints, shall reject unauthenticated requests with an HTTP 401 response.                                                                                                                     | On the protected routes, this will serve as the primary authorization and token validation system. | OWSAP ZAP                      | 100% 401 / 100% 401                                               |
| NFR-1.2 | User credentials shall never be stored in plain text by the system. Sessions shall be established with expiring access tokens with a lifetime of at most 60 minutes; an expired session shall require re-authentication or a silent token refresh before any protected resource is served. | This makes use of both the expired access token + the identity provider.                           | Live system test + AWS Cognito | 0 plaintext text; <= 60 minutes / 0 plaintext text; <= 60 minutes |
| NFR-1.3 | Multi-factor authentication (TOTP) shall be enforced on user accounts.                                                                                                                                                                                                                     | This is the second-factor authentication for TOTP.                                                 | Live Login Test                | Valid TOTP / Valid TOTP                                           |
| NFR-1.4 | All data in transit shall be encrypted with TLS;                                                                                                                                                                                                                                           | This guaranties both the encryption of the production data + HTTPS/TLS termination.                | OpenSSL                        | TLS / TLS                                                         |
| NFR 1.5 | All portfolio data shall be scoped to the owning user at the data-access level: 0                                                                                                                                                                                                          | User ID checks of all the users.                                                                   | OWSAP                          | 0 cross-user access / 0 cross-user access                         |

|         |                                                                                                                                                         |                                                 |            |                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------|------------|-------------------------------------------|
|         | instances of cross-user data exposure.                                                                                                                  |                                                 |            |                                           |
| NFR-1.6 | All portfolio data shall be scoped to the owning user at the data-access level: 0 instances of cross-user data exposure, verified by integration tests. | Owner-scoped authorization + data access checks | GitLeaks   | 0 exposure / 0 exposure                   |
| NFR-1.7 | All personal data handling shall comply with POPIA.                                                                                                     | Data minimization + access control              | OWSAP ZAP  | All defined controls passed for the POPIA |
| NFR-2.1 | API endpoints shall respond within 500 ms at the 85th percentile under normal load.                                                                     | Fast API + database processing                  | K6         | <= 500 ms / <= 500 ms                     |
| NFR-2.2 | Initial page load shall complete within 10 seconds on a standard broadband connection.                                                                  | Fast frontend + assets loading                  | Lighthouse | <= 10 s / <= 10 s                         |
| NFR-2.3 | Dashboard visualisations shall render within 6 seconds                                                                                                  | Fast dashboard data loading                     | Lighthouse | <= 6 s / <= 6 s                           |

|         |                                                                                                                                                                                                                                    |                                              |               |                                                |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|---------------|------------------------------------------------|
| NFR-2.4 | Market data shall be served through a cache-aside store: a request for an instrument that is already cached shall be answered with zero calls to the upstream provider, keeping repeat dashboard loads within provider rate limits | Cache market data to avoid extra API calls   | Live API Test | 0 upstream calls / 0 upstream calls            |
| NFR-3.1 | The failure of any single external data source (market data, news, AI model) shall degrade only its own feature; the remainder of the system shall stay fully functional.                                                          | The failure isolation + graceful degradation | Pytest        | Other features in the system remain functional |
| NFR-3.2 | Show a valid error message if the AI feature is not working                                                                                                                                                                        | Error handling for the ai service            | Pytest        | Valid error shown / Valid error shown /        |
| NFR-3.3 | A failed fetch for one holding's market data shall not prevent the dashboard from rendering the remaining holdings.                                                                                                                | Error isolation                              | Pytest        | The remaining holders rendering                |
| NFR-4.1 | The backend shall support at least 50 concurrent users without performance degradation beyond the NFR-2 targets.                                                                                                                   | Concurrent User Handling                     | K6            | 50 concurrent users/ 100% passed               |

|         |                                                                                                                                                          |                                                          |                                     |                                                                                                     |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------------------------------------------|
| NFR-4.2 | Send 100 simultaneous API Requests to EquityLens                                                                                                         | High Request Handling                                    | K6                                  | 100 simultaneous request/ 100% passed                                                               |
| NFR-4.3 | The system shall accommodate a 200% workload increase without major architectural changes and with no more than a 10% performance decrease.              | Workload Scaling Support                                 | K6                                  | Test passed after increased the workload by 200% / Test passed after increased the workload by 200% |
| NFR-5.1 | A new user shall be able to register, import a portfolio and reach a populated dashboard within 5 minutes of first use.                                  | Clear validation feedback for the user + easy user setup | Live system Testing                 | <= 5 minutes / <= 5 minutes                                                                         |
| NFR-5.2 | The interface shall meet WCAG 2.1 AA: all interactive elements keyboard-navigable with visible focus states, and text contrast ratios of at least 4.5:1. | Accessible UI Design                                     | Live system tests + chrome DevTools | Keyboard + visible focus + >= 4.5:1 > Paased + >= 4.5:1                                             |

|         |                                                                                                                                                                           |                                                                            |                   |                                           |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------|-------------------------------------------|
| NFR-5.3 | Every financial indicator shall have a plain-language explanation accessible from where the indicator is displayed, so no finance training is required to use the system. | Explanations of each indicator + Contextual help                           | Live system tests | 100% indicators / 100% indicators         |
| NFR-5.4 | The interface shall be fully responsive across desktop (1280px+), tablet (768-1279px) and mobile (320-767px) with no horizontal scrolling on mobile.                      | Responsive layout + it can be used in mobile phones + flexible grid design | Playwright        | 0 horizontal scroll / 0 horizontal scroll |

